

Temperature Uncertainty Viewer

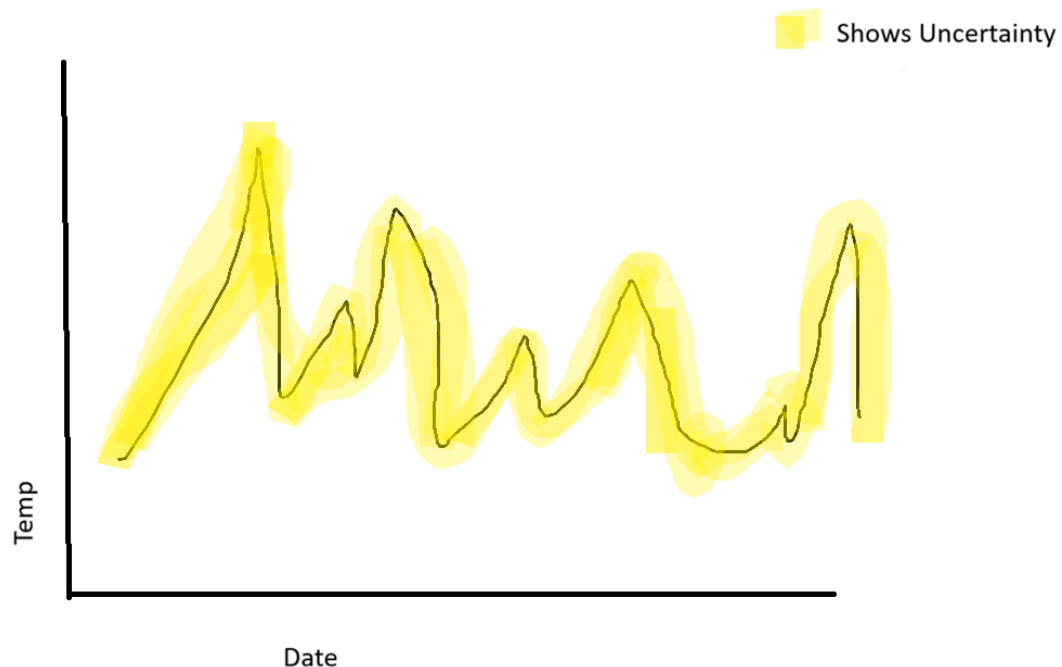
Dataset: <https://www.kaggle.com/datasets/prasad22/weather-data>

Controls:

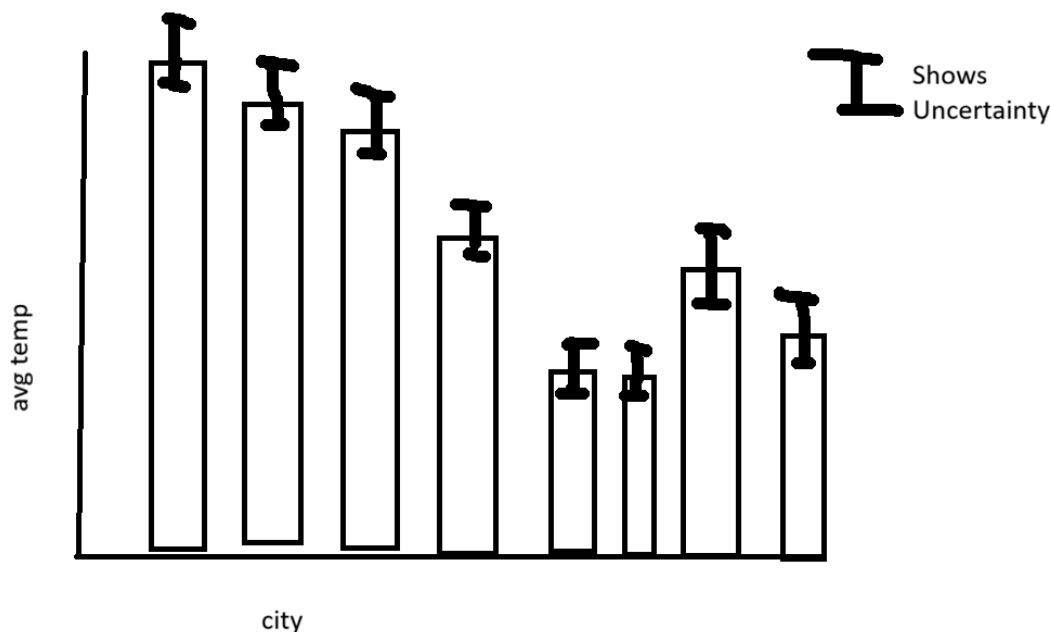
[Dropdown: Select City] [Toggle Uncertainty]

Main Chart: Daily Temperature with Uncertainty

- Size: 800px x 400px
- X-axis: Date
- Y-axis: Temperature (°C)
- Line: Daily average temperature
- Shaded area: Min–Max temperature range
- Interaction:
 - Hover over the chart to show a tooltip with:
 - City
 - Average temperature
 - Min-Max temperature range
- Use Case Example:
 - Identify the day with the largest temperature uncertainty for a selected city
- Toggle Button:
 - Show/Hide the uncertainty area (shaded region)



Second Chart: Bar Graph – Average Temperature per City



- Size: 700px x 350px
- X-axis: City
- Y-axis: Average temperature (°C)
- Bars: Height represents the city's average temperature
- Interaction:
 - Hover over bars to show tooltip with:
 - City name
 - Average temperature
- Use Case Example:
 - Compare average temperatures across all cities to find the hottest city

Visual Encoding Summary:

Element	Encoding
Line (Main Chart)	Blue line = daily avg temp
Shaded Area (Main Chart)	Light blue = min-max temp range
Bars (Bar Chart)	Orange bars = city average temp
Tooltip	Shows detailed numeric info

Interactions Summary:

1. Hover on Main Chart → Tooltip + highlight data (daily temp + uncertainty)
2. Hover on Bar Chart → Tooltip (average temp per city)
3. Dropdown → Select city for Main Chart
4. Toggle Button → Show/Hide uncertainty area on Main Chart

Analysis Tasks:**Task 1:**

Identify which periods in February showed the highest temperature uncertainty for a selected city and determine whether those periods corresponded to major temperature changes.

Task 2:

Compare cities to determine which locations maintained the most stable temperatures throughout February and which experienced the greatest variability.